Core practical – Preparing copper sulfate

Pearson Edexcel GCSE (9-1) Combined Science / Chemistry

Name _____ **Class** _____ **Date** _____

Practical Objective

To prepare a sample of pure, dry, hydrated copper sulfate crystals starting from copper(II) oxide.

Apparatus

- eye protection
- 100 cm^3 conical flask
- 100 cm^3 beaker
- Bunsen burner
- gauze and tripod
- heat mat
- Petri dish or watch glass
- 100 cm^3 measuring cylinder
- evaporating basin
- spatula
- stirring rod
- clamp, stand and boss
- filter funnel
- filter paper
- tongs
- water bath (set at 50 °C)
- dilute sulfuric acid
- copper(II) oxide

Safety

- Wear eye protection at all times.

Your teacher may watch to see if you can:

- safely and correctly use apparatus.

Method

A Pour about 20 cm^3 of dilute sulfuric acid into a conical flask.

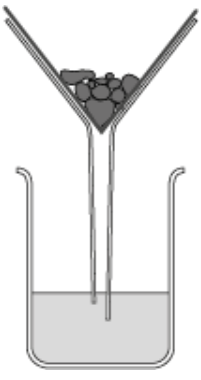
B Place the conical flask into a water bath at 50 °C and heat for 3–4 minutes to allow the acid to heat up.

C Use the spatula to add a little copper(II) oxide to the acid and stir or swirl the contents of the flask.

D Keep repeating step **C** until the black powder does not disappear after stirring. (This makes sure the copper(II) oxide is in excess.)

E Return the mixture to the water bath for a few minutes (to make sure there is no acid left).

F Filter the mixture into a beaker and pour into an evaporating basin.



*Step **F***

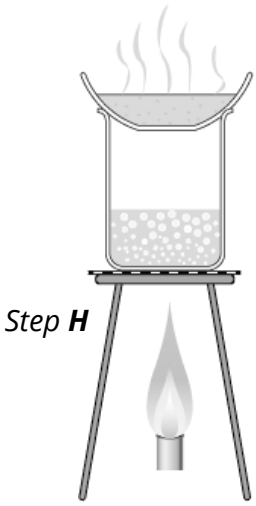
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G Place the evaporating basin on top of a beaker half full of water. Heat the beaker, evaporating basin and contents using a Bunsen burner on a blue flame.

H Heat until about half of the water has evaporated. Then allow the evaporating basin to cool.

I When cool, transfer the solution to a Petri dish or watch glass and leave for a few days to allow the water to evaporate.

J Observe the shape and colour of the copper sulfate crystals formed.



*Step **H***

Recording your results

1 Describe the colour, shape and size of the copper sulfate crystals produced.

_____

_____

2 Describe the appearance of:

a the sulfuric acid

_____

b the copper(II) oxide

_____

c the solution at the end of the reaction.

_____

Considering your results

3 Write a word equation to show the reaction you have carried out.

_____

4 State why you need to be sure excess copper(II) oxide is added in step **D**.

_____

_____

5 What would happen in step **E** if there was still some acid left?

_____

_____

6 Name the substance left in the filter paper in step **F**.

_____

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7 What is dissolved in the solution that went through the filter paper?

_____

8 Explain why this is an example of a neutralisation reaction.

_____

_____

_____

_____

9 What substance acts as a base in this reaction?

_____

10 Write a symbol equation to show the reaction you have carried out. Include the state symbols. Use your answer to question **3** to help you.

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